VIP

Vasoactive Intestinal Peptide | 10mg x 10 Vials

Pulmonary | Immune Modulation | Neuroprotection | Anti-Inflammatory

Purity: 99.8% | Endotoxin-Free | Research Use Only

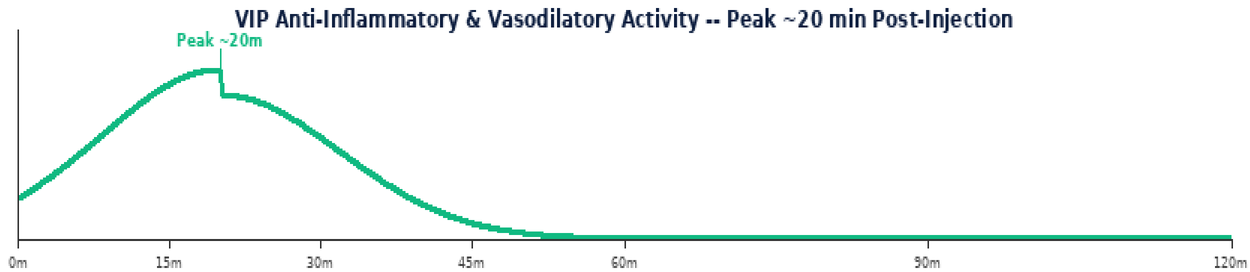
VIP (Vasoactive Intestinal Peptide) is a 28-amino acid neuropeptide produced throughout the nervous system and gastrointestinal tract. It is one of the most potent endogenous anti-inflammatory and immunomodulatory peptides known, acting via VPAC1 and VPAC2 receptors. Research applications span pulmonary arterial hypertension (inhaled), inflammatory disease, autoimmunity, CIRS/MCAS, neuroprotection, and circadian rhythm regulation.

VIP Vial Comparison

10mg
10.0mg/mL

Parameter	Specification
Class	Endogenous 28-AA neuropeptide -- VPAC1/VPAC2 receptor agonist
Vial Size	10mg x 10 vials | Purity 99.8%
Reconstitution	1 mL BAC Water -> 10mg/mL (10,000 mcg/mL)
Standard Dose	500-5,000 mcg per injection SubQ | inhaled for pulmonary protocols
Routes	SubQ (systemic) | Inhaled/nebulised (pulmonary) | IM
Frequency	1-2x daily SubQ | inhaled 2-3x daily | no fasting required
Cycle & Storage	4-12 weeks on, 4 off | 2-8 deg C | use within 28 days

MECHANISM OF ACTION & RESEARCH BENEFITS

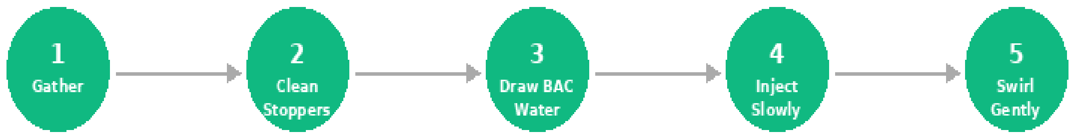


Pathway	Mechanism	Benefit
VPAC1/VPAC2 Anti-Inflammatory	Binds VPAC1 (ubiquitous) and VPAC2 (CNS, lung, immune) receptors; activates cAMP/PKA signalling; potently suppresses NF-kB; reduces TNF-alpha, IL-6, IL-1beta, IFN-gamma; promotes regulatory T-cell (Treg) differentiation; one of the most potent endogenous anti-inflammatory agents known	Inflammatory disease; autoimmunity; CIRS/MCAS research; cytokine storm models
Pulmonary Vasodilation	Potent pulmonary vasodilator via VPAC2 on vascular smooth muscle; reduces pulmonary vascular resistance; inhaled VIP used in clinical trials for pulmonary arterial hypertension (PAH); bronchodilatory effects in airway smooth muscle	PAH research; pulmonary vascular disease; respiratory research
Neuroprotection & Circadian	Produced in suprachiasmatic nucleus (SCN) -- circadian pacemaker; neuroprotective via anti-apoptotic signalling; promotes neurotrophin release; GI motility regulation; potential CIRS/mold illness research applications	Circadian regulation; neuroprotection; CIRS; neuroimmune research

Benefit	Context
Most Potent Endogenous Anti-Inflammatory	VIP suppresses the full cytokine cascade (NF-kB, TNF-alpha, IL-6, IL-1beta) and promotes Treg differentiation -- among the most potent endogenous immunomodulators.
Inhaled Pulmonary Route	Inhaled/nebulised VIP delivers targeted pulmonary vasodilation and bronchodilation for PAH and airway research without systemic hypotension.
CIRS / MCAS Research	VIP deficiency has been documented in CIRS (Chronic Inflammatory Response Syndrome) from mold/biotoxin exposure. Replacement is an active research area.

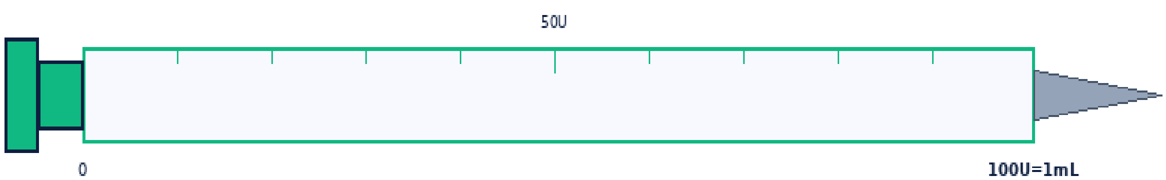
RECONSTITUTION & DOSE CALCULATION

All vials use 1 mL BAC Water. Room temperature. Swirl gently -- never shake.



1 mL BAC Water per vial. Roll gently -- NEVER shake.

Step	Action	Detail
1	Gather	Vial, BAC Water, insulin syringe (29-31G), swabs, sharps.
2	Clean stoppers	Wipe both stoppers. Air-dry 10-15 sec.
3	Draw 1mL BAC	Exactly 1 mL. Expel air bubbles.
4	Inject into vial	Aim at GLASS WALL. Slowly. Never onto powder.
5	Swirl & label	Roll 20-30 sec. Clear within 60 sec. Never shake. Label date. 2-8 deg C. Use within 28 days.



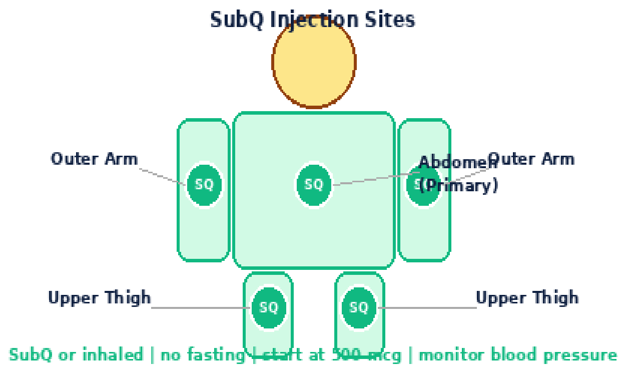
1mL Insulin Syringe | 29-31G | SubQ

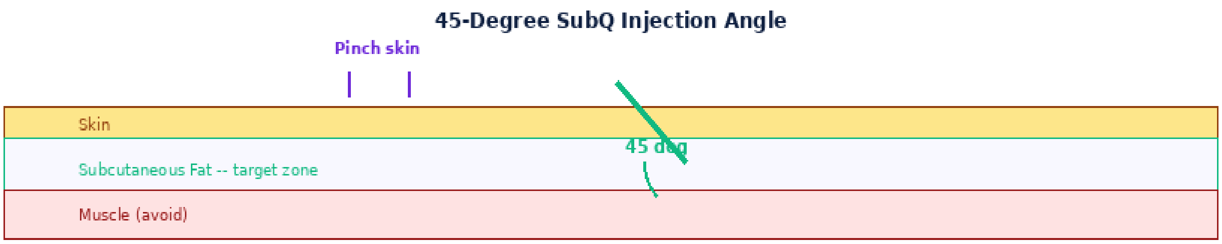
Formula: Vol(mL)=Dose(mcg)/Conc(mcg/mL) **| Units:** Vol x 100

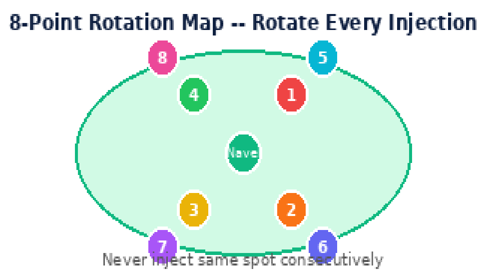
10mg x 10 -- 1mL BAC -> 10.0mg/mL (10,000mcg/mL)

Dose(mcg)	500mcg	1000mcg	2000mcg	5000mcg
Vol(mL)	0.0500	0.1000	0.2000	0.5000
Units	5.0U	10.0U	20.0U	50.0U
Doses/Vial	20	10	5	2

INJECTION TECHNIQUE, PROTOCOLS & GOLDEN RULES







Step	Action	Detail
1	Select route	SubQ (systemic): 29-31G abdomen. Inhaled: dilute in saline for nebuliser.
2	Draw/prepare	SubQ: draw volume per dose table. Inhaled: dilute 500-2,000 mcg in 3mL saline for nebuliser.
3	Administer	SubQ: pinch, 45 deg, inject slowly. Inhaled: nebulise over 5-10 min, breathe normally.
4	Rotate & dispose	Rotate SubQ sites. 8-point rotation. Sharps container.

Protocol	Dose	Frequency	Duration	Notes
Systemic SubQ	500-2,000 mcg	1-2x daily	4-12 wk 4 off	No fasting. Anti-inflammatory systemic protocol.
Inhaled (Pulmonary)	500-2,000 mcg in 3mL saline	2-3x daily	4-12 wk 4 off	Nebulise over 5-10 min. PAH / respiratory research.
CIRS/MCAS Research	1,000-5,000 mcg	1-2x daily	Per protocol w/ monitoring	Under medical supervision. Monitor labs.

Side Effect	Likelihood	Management
Flushing / transient hypotension	Dose-related	VIP is vasodilatory. Start low. Inject slowly. Usually transient.
Nausea	Occasional	GI motility effect. Usually mild and transient.

Injection site reaction	Common	Rotate sites. Fresh needle.
Rapid heart rate (tachycardia)	High dose	VIP is chronotropic. Monitor heart rate. Reduce dose.

Contraindications: Cancer history, pregnancy, under 18.

1. 1 mL BAC Water -> 10mg/mL (10,000 mcg/mL).
2. No fasting required. Inject any time.
3. Start at 500 mcg to assess vasodilatory tolerance before escalating.
4. Flushing and transient blood pressure drop are expected at higher doses -- start low.
5. Inhaled route for pulmonary protocols: dilute 500-2,000 mcg in 3mL 0.9% saline, nebulise.
6. CIRS/MCAS protocols require medical supervision and regular laboratory monitoring.
7. VIP is vasodilatory and chronotropic -- monitor blood pressure and heart rate.
8. Rotate SubQ sites through 8-point abdomen map. Store 2-8 deg C, use within 28 days.
9. Contraindicated in severe hypotension and known VIP hypersensitivity.
10. Not for use during pregnancy, lactation, or in individuals under 18.